

with the exception that if the eigenvalue μ^2 vanishes then $\sum_B c_B t_B \phi_0 = 0$, so that the eigenvectors u and v are absent. The massless particles associated with the v eigenvectors are Goldstone bosons, which are eliminated from the physical spectrum by the unitarity gauge condition (27.4.14). In addition to these mass eigenstates, there is another set that are orthogonal to all the u s and v s and that therefore take the form

$$w_{\pm} = \begin{bmatrix} \zeta \\ \pm \zeta^* \end{bmatrix}, \quad (27.4.24)$$

where

$$\sum_n (t_A \phi_0)_n^* \zeta_n = 0. \quad (27.4.25)$$

Eq. (27.4.19) shows that the space of ζ s satisfying Eq. (27.4.25) is invariant under multiplication with the Hermitian matrix $\mathcal{M}^\dagger \mathcal{M}$, so it is spanned by the eigenvectors of this matrix, satisfying

$$\mathcal{M}^\dagger \mathcal{M} \zeta = m^2 \zeta \quad (27.4.26)$$

with m^2 a set of real positive (or zero) eigenvalues. Eq. (27.4.26) and its complex conjugate together with Eq. (27.4.25) show that $w_{\pm}$ are eigenvectors of M_0^2 with eigenvalues m^2 :

$$M_0^2 w_{\pm} = m^2 w_{\pm}. \quad (27.4.27)$$

We thus have *two* self-charge-conjugate spinless bosons of each mass m satisfying Eq. (27.4.27), and one self-charge-conjugate spinless boson of each non-zero mass μ satisfying Eq. (27.4.23).

Spin 1/2

The fermion masses arise from the non-derivative terms in Eq. (27.4.8) that are of second order in the fermion fields ψ_n and λ_A :

$$\mathcal{L}_{1/2} = -\frac{1}{2} \sum_{nm} \mathcal{M}_{nm} (\psi_{nL}^T \epsilon \psi_{mL}) - i\sqrt{2} \sum_{Am} (t_A \phi_0)_m^* (\lambda_{LA}^T \epsilon \psi_{mL}) + \text{H.c.} \quad (27.4.28)$$

We saw in Section 26.4 that if the fermion mass term in the Lagrangian for a column χ of Majorana spinor fields is put in the form

$$\mathcal{L}_{1/2} = -\frac{1}{2} (\chi_L^T \epsilon M \chi_L) + \text{H.c.}, \quad (27.4.29)$$

then the fermion squared masses are the eigenvalues of the Hermitian matrix $M^\dagger M$. Eq. (27.4.28) gives the elements of the matrix M here as

$$M_{nm} = \mathcal{M}_{nm}, \quad M_{nA} = M_{An} = i\sqrt{2} (t_A \phi_0)_n^*, \quad M_{AB} = 0, \quad (27.4.30)$$